

CS 4/5782 2-Page Project Summary Report

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Spring 2026 Team 14 🐙 GitHub Repository <https://github.com/kalvin1019/GAT-CS5782-Final-Project>

1 Introduction

Attention [1] has been popularized among different neuron network structures and in the Petar Velickovi'c presented the idea of combining graphic neuron networks and attention mechanism and named it as Graphic Attention Networks (GATs) [2]. Futhermore, Shaked Brody presents the limitations of the naive version of GAT - the static attention mechanism and instead used dynamic attention mechanism and named it as GATv2 [3].

2 Replication

For CORA, the original paper (Graphic Attention Networks) used 140 training sample, 500 validation sample, and 1000 remaining testing sample. We downloaded the raw files from the github link provided in the paper and ran in Google Colab. Since the Tensor Flow version was old, we rewrote parts of the code and successfully replicated the result with a testing accuracy 82.6%. This fits into the expected accuracy with $83.0\% \pm 0.7\%$.

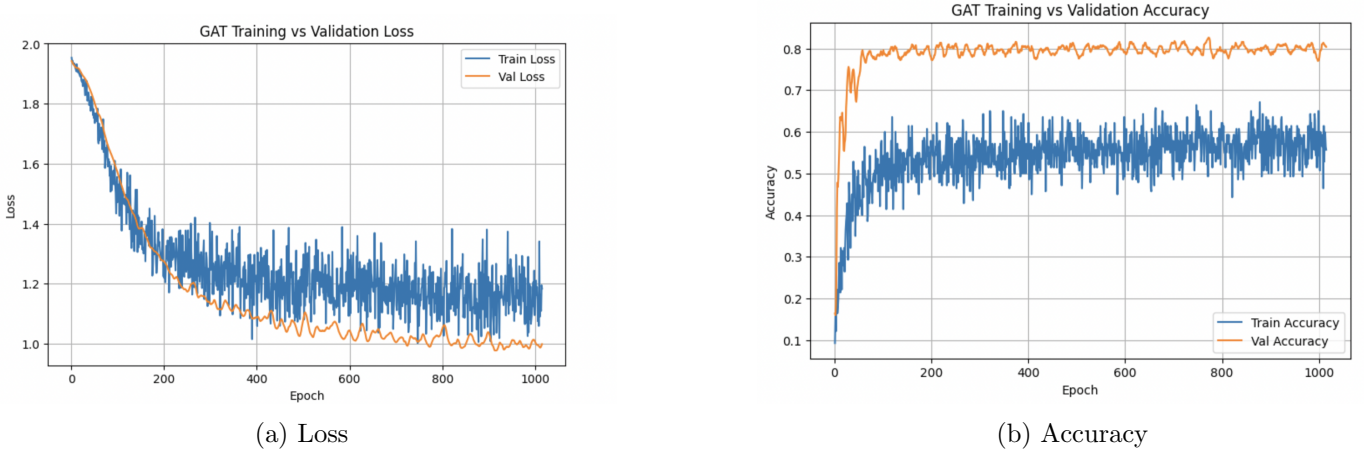


Figure 1: Cora Replication - GAT Training Results

3 Methodology

We reproduced the original Graph Attention Network (GAT) on the Cora citation dataset for semi-supervised node classification. The original implementation was based on TensorFlow 1.x, so we adapted the code for compatibility with modern Colab and TensorFlow environments while preserving the original training setup.

The baseline GAT computes attention scores between neighboring nodes using node features:

$$e_{ij} = \text{LeakyReLU} (a^T [Wh_i \parallel Wh_j])$$

The original GAT computes attention mainly from node feature similarity and does not explicitly utilize richer graph topology information. However, in citation networks such as Cora, structurally related nodes often share similar semantic labels. To better capture these relationships, we introduce a structural feature vector ϕ_{ij} into the attention computation.

$$e_{ij} = \text{LeakyReLU} (a^T [Wh_i \parallel Wh_j \parallel \phi_{ij}])$$

where

$$\phi_{ij} = \left[\frac{\deg(j)}{\deg_{\max}}, \frac{|N_i \cap N_j|}{|N_i \cup N_j|}, \sum_{k \in N_i \cap N_j} \frac{1}{\log(\deg(k))} \right]$$

The three terms correspond to degree centrality, Jaccard similarity, and Adamic-Adar similarity.

We also explored an alternative projection strategy GATv2 by applying the projection matrix after feature concatenation:

$$e_{ij} = a^\top \text{LeakyReLU}(W[h_i \parallel h_j])$$

and further combined it with the structural feature vector:

$$e_{ij} = a^\top \text{LeakyReLU}((W[h_i \parallel h_j] \parallel \phi_{ij}))$$

We compared four variants: the original GATv1 baseline, GATv1 with ϕ_{ij} , GATv2 baseline, and GATv2 with ϕ_{ij} . All experiments were evaluated using node classification accuracy on the Cora test set.

4 Results and Analysis

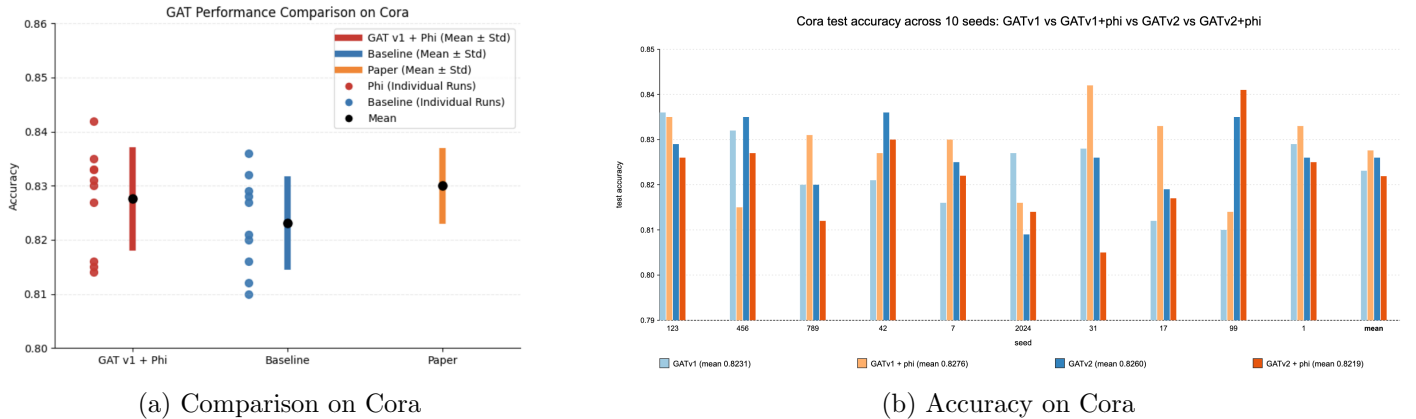


Figure 2: Comparison

Figure 2 shows that adding the structural feature vector ϕ_{ij} slightly improves the average test accuracy compared to the baseline GAT while maintaining stable convergence during training. The added structural information only introduces lightweight neighborhood-based computations with approximately $O(E \cdot d)$ complexity, so the overall computational and memory overhead remains small.

For theoretical justification, we argue that the original GAT architecture is a special case of our GAT + ϕ architecture with $\mathbf{a}_\phi = \mathbf{0}$. Also, for some data sets, the usual assumption is the homophily graph with $\Pr(y_i = y_j \mid \text{high structural similarity}) > \Pr(y_i = y_j)$, which is why the results are justified in datasets like Cora, where structure-dependent weights better preserve label-smooth signals on the graph.

5 Reflections

Through this project, we gained a better understanding of graph attention mechanisms and graph neural networks. Our experiments show that adding structural information into the attention computation can provide small but consistent improvements on the Cora dataset with limited computational overhead. In future work, we plan to evaluate the method on larger datasets and explore more advanced graph attention architectures.

References

- [1] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In I. Guyon, U. Von Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017.
- [2] Petar Veličković, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. Graph attention networks. In *International Conference on Learning Representations*, 2018.
- [3] Shaked Brody, Uri Alon, and Eran Yahav. How attentive are graph attention networks? In *International Conference on Learning Representations*, 2022.